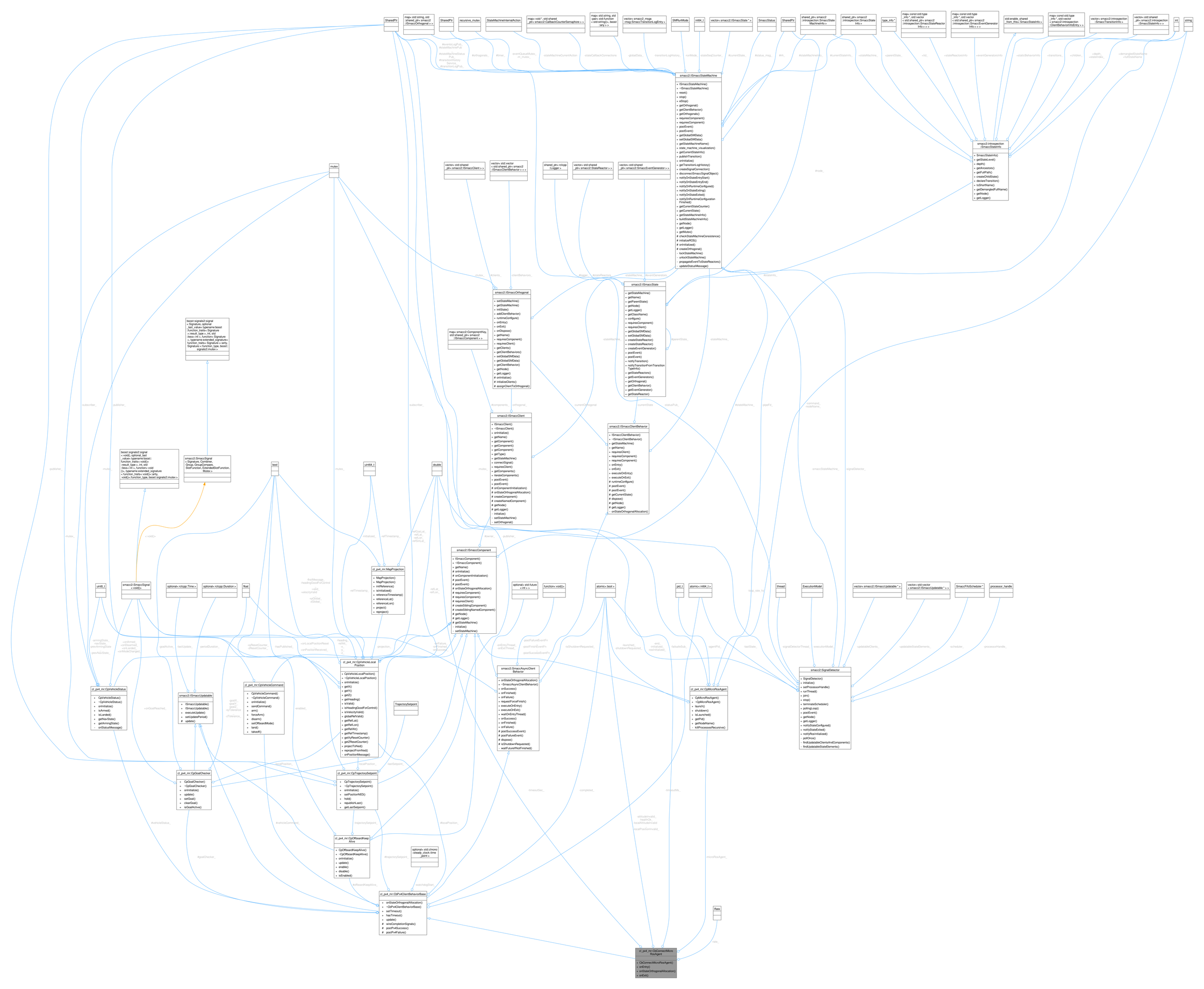
SharedPtr
map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >
SharedPtr
recursive_mutex
StateMachineInternalAction
map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std ::pair< std::function < std::string()>, boost ::any > >
vector< smacc2_msgs ::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
vector< smacc2::ISmaccState * >
SmaccStatus
SharedPtr
shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >
shared_ptr< smacc2 ::introspection::SmaccState Info >
type_info *
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccStateReactor Info > > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccEventGenerator Info > > >
std::enable_shared _from_this< SmaccStateInfo >
map< const std::type _info *, std::vector < smacc2::introspection ::ClientBehaviorInfoEntry > >
vector< smacc2::introspection ::SmaccTransitionInfo >
vector< std::shared _ptr< smacc2::introspection ::SmaccStateInfo > >
int
string
#eventsLogPub_
#stateMachinePub_
#stateMachineStatus Pub_
#transitionHistory Service_
#transitionLogPub_
#orthogonals_
#timer_
-eventQueueMutex_ -m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections_
-globalData_
-transitionLogHistory_
-runMode_
-stateSeqCounter_
-currentState_
-status_msg_
-lm_
-stateMachineInfo_
-currentStateInfo_
-stateMachine_
-parentState_
-tid_
-stateReactorsInfo
-eventGeneratorsInfo
-staticBehaviorInfo
-transitions_
-children_
-depth_ -stateIndex_
-demangledStateName -fullStateName_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
smacc2::introspection ::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
#node_
#stateInfo_
mutex
vector< std::shared _ptr< smacc2::ISmaccClient > >
vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >
shared_ptr< rclcpp ::Logger >
vector< std::shared _ptr< smacc2::StateReactor > >
vector< std::shared _ptr< smacc2::SmaccEventGenerator > >
-mutex_
#clients_
-clientBehaviors_
#logger_
#stateReactors_
-stateMachine_
-eventGenerators_
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
smacc2::ISmaccState
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
-stateMachine_
#parentState_
-stateMachine_
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >
#components_
-orthogonal_
-currentOrthogonal
-currentState_
-statusPub_
#stateMachine_
-pipeFd_
-command_ -nodeName_
boost::signals2::signal < Signature, optional _last_value< typename boost ::function_traits< Signature ::result_type >::type >, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost:: signals2::mutex >
boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()> ::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
smacc2::SmaccSignal < void()>
-subscriber_
publisher_
-mutex_
< void()>
bool
uint64_t
double
-initialized_
-refTimestamp_
-refCosLat_ -refLat_ -refLon_ -refSinLat_
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponents()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
#owner_
publisher_
cl_px4_mr::MapProjection
+ MapProjection()
+ MapProjection()
+ initReference()
+ isInitialized()
+ referenceTimestamp()
+ referenceLat()
+ referenceLon()
+ project()
+ reproject()
-firstMessage_ -headingGoodForControl
-valid_ -velocityValid_
-refTimestamp_
-refLat_ -refLon_
-xyGlobal_ -vGlobal_
optional< std::future < R*t > >
function< void()>
atomic< bool >
pid_t
atomic< int64_t >
-loop_rate_hz_
thread
ExecutionModel
vector< smacc2::ISmaccUpdatable * > >
vector< std::vector < smacc2::ISmaccUpdatable * > > >
SmaccFifoScheduler *
processor_handle
uint8_t
smacc2::SmaccSignal < void()>
optional< rclcpp::Time >
optional< rclcpp::Duration >
float
-armingState_ -navState_ -prevArmingState_ -prevNavState_
-onArmed_ -onDisarmed_ -onLanded_ -onModeChanged_
-goalActive_
-lastUpdate_
-periodDuration_
-xyResetCounter_ -zResetCounter_
-hasPublished_
-onLocalPositionReset_ -onPositionReceived_
-heading_ -refAlt_ -x_ -y_ -z_
-projection_
-onFailure_ -onFinished_ -onSuccess_
-onEntryThread_ -onExitThread_
-postFailureEventFn -postFinishEventFn -postSuccessEventFn
-isShutdownRequested_
-launched_ -shutdownRequested_
-end_ -initialized_ -wasInitialized_
-failsafeSub_
-agentPid_
-lastState_
-signalDetectorThread_
-executionMode_
-updatableClients_
-updatableStateElements_
-scheduler_
-processorHandle_
cl_px4_mr::CpVehicleStatus
+ CpVehicleStatus()
+ ~CpVehicleStatus()
+ onInitialize()
+ isArmed()
+ isLanded()
+ getNavState()
+ getArmingState()
- onStatusMessage()
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()
cl_px4_mr::CpVehicleCommand
+ CpVehicleCommand()
+ ~CpVehicleCommand()
+ onInitialize()
+ sendCommand()
+ arm()
+ forceArm()
+ disarm()
+ setOffboardMode()
+ land()
+ takeoff()
-goalX_ -goalY_ -goalZ_ -xyTolerance_ -zTolerance_
-enabled_
cl_px4_mr::CpVehicleLocal Position
+ CpVehicleLocalPosition()
+ ~CpVehicleLocalPosition()
+ oninitialize()
+ getX()
+ getY()
+ getZ()
+ getHeading()
+ isValid()
+ isHeadingGoodForControl()
+ isVelocityValid()
+ globalRefValid()
+ getRefLat()
+ getRefLon()
+ getRefAlt()
+ getRefTimestamp()
+ getXyResetCounter()
+ getZResetCounter()
+ projectToNed()
+ reprojectFromNed()
- onPositionMessage()
-localPosition_
TrajectorySetpoint
smacc2::SmaccAsyncClient Behavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFuture(fNotFinished)
cl_px4_mr::CpMicroRosAgent
+ CpMicroRosAgent()
+ ~CpMicroRosAgent()
+ launch()
+ shutdown()
+ isLaunched()
+ getPid()
+ getNodeName()
- killProcessesRecursive()
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
-signalDetector_
cl_px4_mr::CpGoalChecker
+ CpGoalChecker()
+ ~CpGoalChecker()
+ onInitialize()
+ update()
+ setGoal()
+ clearGoal()
+ isGoalActive()
-lastSetpoint_
cl_px4_mr::CpTrajectorySetpoint
+ CpTrajectorySetpoint()
+ ~CpTrajectorySetpoint()
+ onInitialize()
+ setPositionNED()
+ hold()
+ republishLast()
+ getLastSetpoint()
-timeoutSec_
-completed_
-timeoutMs_
#vehicleStatus_
#vehicleCommand_
-trajectorySetpoint_
#localPosition_
cl_px4_mr::CpOffboardKeep Alive
+ CpOffboardKeepAlive()
+ ~CpOffboardKeepAlive()
+ onInitialize()
+ update()
+ enable()
+ disable()
+ isEnabled()
optional< std::chrono ::steady_clock::time _point >
-attitudeInvalid_ -healthOk_ -localAttitudeInvalid -localPositionInvalid_
#goalChecker_
#trajectorySetpoint_
#offboardKeepAlive_
-watchdogStart_
cl_px4_mr::CbPx4ClientBehaviorBase
+ onStateOrthogonalAllocation()
+ ~CbPx4ClientBehaviorBase()
+ setTimeout()
+ hasTimeout()
+ update()
# wireCompletionSignals()
# postPx4Success()
# postPx4Failure()
-microRosAgent_
Rate
-rate_
cl_px4_mr::CbConnectMicro RosAgent
+ CbConnectMicroRosAgent()
+ onEntry()
+ onStateOrthogonalAllocation()
+ onExit()